ML24-02-Gait PatternR ecognition

Literature Review

Abishek Senthilkumar
Bruna Teixeira
Fedor Kotiaev

31. Januar 2025

1. Literature Review Article: 1
2. Literature Review Article: 2
3. Literature Review Article: 3
4. Literature Review Article: 4
5. Literature Review Article: 5
6. Literature Review Article: 6
7. Literature Review Article: 7
8. Literature Review Article: 8

9 Literature Review Article: 9

10 Literature Review Article: 10

11 Literature Review Article: 11

12 Literature Review Article: 12

13 Literature Review Article: 13

14 Literature Review Article: 14

15 Literature Review Article: 15

16 Literature Review Article: 16

17 Literature Review Article: 17

18 Literature Review Article: 18

19 Literature Review Article: 19

20 Literature Review Article: 20

21 Literature Review Article: 21

22 Literature Review Article: 22

23 Literature Review Article: 23

24 Literature Review Article: 24

25 Literature Review Article: 25

26 Literature Review Article: 26

27 Literature Review Article: 27

28 Literature Review Article: 28

29 Literature Review Article: 29

30 Literature Review Article: 30

31 Literature Review Article: 31

32 Literature Review Article: 32

33 Literature Review Article: 33

34 Literature Review Article: 34

35 Literature Review Article: 35

36 Literature Review Article: 36

37 Literature Review Article: 37

38 Literature Review Article: 38

39 Literature Review Article: 39

40 Literature Review Article: 40

41 Literature Review Article: 41

42 Literature Review Article: 42

43 Literature Review Article: 43

44 References

Literature Review Article: 1

Topic : Keyword Spotting System with Nano 33 BLE Sense Using Embedded Machine Learning Approach I

Description: This study presents a TinyML model implemented on the Arduino Nano 33 BLE Sense for keyword spotting. The model uses Mel-Frequency Cepstral Coefficients (MFCC) and Convolutional Neural Networks (CNN) to achieve an accuracy of 84.51

Keywords: Edge Impulse, Keyword Spotting, TinyML, MFCC, CNN.

Topic : Keyword Spotting System with Nano 33 BLE Sense Using Embedded Machine Learning Approach II

Conclusion: The study successfully demonstrates the implementation of a keyword spotting system on the Arduino Nano 33 BLE Sense using TinyML. The model achieves competitive accuracy despite the hardware limitations, highlighting the potential of embedded machine learning for real-time applications. Future work could explore further optimization techniques and the integration of additional sensors to enhance performance. This research contributes to the growing field of TinyML and its applications in edge computing[AA23].

Literature Review Article: 2

Topic : ADC Conversion Basics I

Description: This application note by STMicroelectronics provides a comprehensive introduction to Analog-to-Digital Converter (ADC) basics. It covers the fundamental principles of ADC operation, including sampling, quantization, and resolution, and discusses various ADC architectures and their applications in embedded systems.

Keywords: ADC, Analog-to-Digital Conversion, Sampling, Quantization, Resolution.

Topic : ADC Conversion Basics II

Conclusion: The document serves as a valuable resource for understanding ADC technology and its role in embedded systems. It highlights the importance of selecting the appropriate ADC architecture based on application requirements, such as speed, accuracy, and power consumption. This foundational knowledge is essential for designing efficient and reliable embedded systems[STM24].

Literature Review Article: 3

Topic : Adafruit LSM9DS1 Library I

Description: The Adafruit LSM9DS1 library provides an open-source implementation for interfacing with the LSM9DS1 IMU sensor. The library supports accelerometer, gyroscope, and magnetometer functionalities, enabling developers to easily integrate motion sensing capabilities into their projects.

Keywords: LSM9DS1, IMU, Accelerometer, Gyroscope, Magnetometer, Adafruit.

Topic : Adafruit LSM9DS1 Library II

Conclusion: The Adafruit LSM9DS1 library simplifies the process of working with the LSM9DS1 sensor, making it accessible for hobbyists and professionals alike. Its comprehensive documentation and example code facilitate rapid prototyping and development. This library is a valuable tool for projects requiring motion sensing and orientation tracking[Ada25].

Literature Review Article: 4

Topic : Getting Started with IMU and Accelerometer on Arduino Nano 33 BLE Sense I

Description: This tutorial by Arduino Documentation provides a step-by-step guide to using the IMU and accelerometer on the Arduino Nano 33 BLE Sense. It covers sensor initialization, data acquisition, and basic signal processing techniques for motion detection.

Keywords: IMU, Accelerometer, Arduino Nano 33 BLE Sense, Motion Detection.

Topic : Getting Started with IMU and Accelerometer on Arduino Nano 33 BLE Sense II

Conclusion: The tutorial is an excellent resource for beginners looking to explore motion sensing with the Arduino Nano 33 BLE Sense. It provides practical examples and clear explanations, enabling users to quickly implement motion-based applications. This guide lays the foundation for more advanced projects involving IMU and accelerometer data[Alu24].

Literature Review Article: 5

Topic : Arduino Guide I

Description: The Arduino Guide is a comprehensive resource for getting started with Arduino development. It covers hardware setup, software installation, and basic programming concepts, making it an essential reference for beginners and experienced developers alike.

Keywords: Arduino, Hardware Setup, Software Installation, Programming.

Topic : Arduino Guide II

Conclusion: The Arduino Guide is a valuable resource for anyone starting with Arduino development. It provides clear instructions and examples, enabling users to quickly set up and begin programming their Arduino boards. This guide is a must-read for those new to embedded systems and IoT development[Ard25].

Literature Review Article: 6

Topic : A Difference Between Arduino Nano 33 BLE Sense vs Sense Lite I

Description: This Arduino Forum discussion highlights the differences between the Arduino Nano 33 BLE Sense and the Sense Lite versions. It compares their features, capabilities, and use cases, helping users choose the right board for their projects.

Keywords: Arduino Nano 33 BLE Sense, Sense Lite, Comparison, Features.

Topic : A Difference Between Arduino Nano 33 BLE Sense vs Sense Lite II

Conclusion: The discussion provides valuable insights into the differences between the two Arduino boards, helping users make informed decisions based on their project requirements. It emphasizes the importance of understanding hardware specifications and capabilities when selecting development boards[Com23].

Literature Review Article: 7

Topic : Understanding I2C Protocol with Arduino I

Description: This tutorial by Arduino Documentation explains the I2C communication protocol and its implementation on Arduino boards. It covers the basics of I2C, including addressing, data transmission, and error handling, with practical examples.

Keywords: I2C, Communication Protocol, Arduino, Data Transmission.

Topic : Understanding I2C Protocol with Arduino II

Conclusion: The tutorial is a valuable resource for understanding and implementing I2C communication on Arduino boards. It provides clear explanations and practical examples, making it accessible for beginners and experienced developers alike. This guide is essential for projects requiring inter-device communication[Ard24b].

Literature Review Article: 8

Topic : The Arduino Software (IDE) - Starting Guide I

Description: This guide provides an introduction to the Arduino Integrated Development Environment (IDE). It covers installation, configuration, and basic usage, enabling users to write, compile, and upload code to Arduino boards.

Keywords: Arduino IDE, Software Development, Code Compilation, Uploading.

Topic : The Arduino Software (IDE) - Starting Guide II

Conclusion: The Arduino IDE Starting Guide is an essential resource for anyone new to Arduino development. It provides step-by-step instructions for setting up and using the IDE, making it easy to get started with Arduino programming. This guide is a must-read for beginners[Ard].

Literature Review Article: 9

Topic : Arduino_LSM9DS1 Library Documentation I

Description: The Arduino_LSM9DS1 library documentation provides detailed information on using the LSM9DS1 IMU sensor with Arduino boards. It includes examples for reading accelerometer, gyroscope, and magnetometer data, as well as configuring the sensor for specific use cases.

Keywords: LSM9DS1, IMU, Arduino, Sensor Configuration.

Topic : Arduino_LSM9DS1 Library Documentation II

Conclusion: The Arduino_LSM9DS1 library documentation is a comprehensive resource for working with the LSM9DS1 sensor. It provides clear examples and explanations, enabling users to quickly integrate motion sensing into their projects. This library is a valuable tool for developers working with IMU sensors[Libnd].

Literature Review Article: 10

Topic : Arduino Nano 33 BLE Sense - Technical Reference and Overview I

Description: This technical reference provides an overview of the Arduino Nano 33 BLE Sense, including its features, specifications, and use cases. It also includes tutorials and examples for using the board's built-in sensors and connectivity options.

Keywords: Arduino Nano 33 BLE Sense, Technical Reference, Sensors, Connectivity.

Topic : Arduino Nano 33 BLE Sense - Technical Reference and Overview II

Conclusion: The Arduino Nano 33 BLE Sense technical reference is a valuable resource for understanding the capabilities of this powerful development board. It provides detailed information and practical examples, making it an essential guide for developers working on IoT and embedded systems projects[Doc25].

Literature Review Article: 11

Topic : Arduino Nano 33 BLE Sense Rev2 Pinout I

Description: This document provides the pinout diagram for the Arduino Nano 33 BLE Sense Rev2, detailing the functions of each pin and its connectivity options. It serves as a quick reference for hardware designers and developers.

Keywords: Arduino Nano 33 BLE Sense, Pinout, Hardware Design, Connectivity.

Topic : Arduino Nano 33 BLE Sense Rev2 Pinout II

Conclusion: The pinout diagram is an essential resource for developers working with the Arduino Nano 33 BLE Sense Rev2. It simplifies the process of designing circuits and interfacing with external components, ensuring efficient and error-free hardware development[Ard24e].

Literature Review Article: 12

Topic : Introduction to PWM on Arduino I

Description: This tutorial introduces Pulse Width Modulation (PWM) and its implementation on Arduino boards. It explains the concept of PWM, its applications, and provides examples for controlling devices such as LEDs and motors.

Keywords: PWM, Pulse Width Modulation, Arduino, Analog Output.

Topic : Introduction to PWM on Arduino II

Conclusion: The tutorial is a valuable resource for understanding and implementing PWM on Arduino boards. It provides clear explanations and practical examples, making it accessible for beginners and experienced developers alike. This guide is essential for projects requiring precise control of analog outputs[Ard24a].

Literature Review Article: 13

Topic : SPI Communication with Arduino I

Description: This tutorial explains the Serial Peripheral Interface (SPI) communication protocol and its implementation on Arduino boards. It covers the basics of SPI, including data transmission, clock synchronization, and device addressing, with practical examples.

Keywords: SPI, Serial Peripheral Interface, Arduino, Data Transmission.

Topic : SPI Communication with Arduino II

Conclusion: The tutorial is a valuable resource for understanding and implementing SPI communication on Arduino boards. It provides clear explanations and practical examples, making it accessible for beginners and experienced developers alike. This guide is essential for projects requiring high-speed data transfer between devices[Ard24b].

Literature Review Article: 14

Topic : UART Communication on Arduino I

Description: This tutorial explains the Universal Asynchronous Receiver-Transmitter (UART) communication protocol and its implementation on Arduino boards. It covers the basics of UART, including data framing, baud rate, and error detection, with practical examples.

Keywords: UART, Serial Communication, Arduino, Data Transmission.

Topic : UART Communication on Arduino II

Conclusion: The tutorial is a valuable resource for understanding and implementing UART communication on Arduino boards. It provides clear explanations and practical examples, making it accessible for beginners and experienced developers alike. This guide is essential for projects requiring reliable serial communication[Ard24a].

Literature Review Article: 15

Topic : Arduino Nano 33 BLE Sense Rev2 Product Reference Manual I

Description: This product reference manual provides detailed technical specifications and usage guidelines for the Arduino Nano 33 BLE Sense Rev2. It includes information on hardware features, sensor integration, and programming examples.

Keywords: Arduino Nano 33 BLE Sense, Product Manual, Hardware Specifications, Sensor Integration.

Topic : Arduino Nano 33 BLE Sense Rev2 Product Reference Manual II

Conclusion: The product reference manual is an essential resource for developers working with the Arduino Nano 33 BLE Sense Rev2. It provides comprehensive information and practical examples, ensuring efficient and effective use of the board's capabilities. This manual is a must-read for anyone developing IoT and embedded systems projects[S.r24].

Literature Review Article: 16

Topic : Arduino Nano 33 BLE Sense Rev2 - Cheat Sheet I

Description: This cheat sheet provides a quick reference for the Arduino Nano 33 BLE Sense Rev2, summarizing its key features, pinout, and usage tips. It is designed to help developers quickly access essential information during project development.

Keywords: Arduino Nano 33 BLE Sense, Cheat Sheet, Quick Reference, Pinout.

Topic : Arduino Nano 33 BLE Sense Rev2 - Cheat Sheet II

Conclusion: The cheat sheet is a valuable resource for developers working with the Arduino Nano 33 BLE Sense Rev2. It provides a concise summary of the board's features and capabilities, enabling quick and efficient project development. This cheat sheet is a handy tool for both beginners and experienced developers[Ard24b].

Literature Review Article: 17

Topic : Arduino Nano 33 BLE Sense Rev2 - Barometric Sensor Tutorial I

Description: This tutorial provides a step-by-step guide to using the barometric sensor on the Arduino Nano 33 BLE Sense Rev2. It covers sensor initialization, data acquisition, and basic signal processing techniques for measuring atmospheric pressure.

Keywords: Barometric Sensor, Arduino Nano 33 BLE Sense, Atmospheric Pressure, Sensor Integration.

Topic : Arduino Nano 33 BLE Sense Rev2 - Barometric Sensor Tutorial II

Conclusion: The tutorial is an excellent resource for developers looking to integrate barometric sensing into their projects. It provides clear explanations and practical examples, enabling users to quickly implement pressure-based applications. This guide lays the foundation for more advanced projects involving environmental sensing[Ard24a].

Literature Review Article: 18

Topic : Use the Built-in Sensors on Nano 33 BLE Sense I

Description: This guide explains how to use the built-in sensors on the Arduino Nano 33 BLE Sense, including the IMU, microphone, and environmental sensors. It provides examples for reading and processing sensor data, enabling developers to create sensor-based applications.

Keywords: Arduino Nano 33 BLE Sense, Built-in Sensors, IMU, Microphone, Environmental Sensors.

Topic : Use the Built-in Sensors on Nano 33 BLE Sense II

Conclusion: The guide is a valuable resource for developers working with the Arduino Nano 33 BLE Sense. It provides practical examples and clear explanations, enabling users to quickly integrate and utilize the board's built-in sensors. This guide is essential for projects requiring multi-sensor data acquisition and processing[Sup24].

Literature Review Article: 19

Topic : The Arduino Software (IDE) I

Description: This guide provides an introduction to the Arduino Integrated Development Environment (IDE). It covers installation, configuration, and basic usage, enabling users to write, compile, and upload code to Arduino boards.

Keywords: Arduino IDE, Software Development, Code Compilation, Uploading.

Topic : The Arduino Software (IDE) II

Conclusion: The Arduino IDE Starting Guide is an essential resource for anyone new to Arduino development. It provides step-by-step instructions for setting up and using the IDE, making it easy to get started with Arduino programming. This guide is a must-read for beginners[Ard24].

Literature Review Article: 20

Topic : Getting Started with Arduino IDE 2: Downloading and Installing I

Description: This guide provides step-by-step instructions for downloading and installing the Arduino IDE 2. It covers the new features and improvements in the IDE, as well as basic usage tips for getting started with Arduino development.

Keywords: Arduino IDE 2, Software Installation, New Features, Arduino Development.

Topic : Getting Started with Arduino IDE 2: Downloading and Installing II

Conclusion: The guide is a valuable resource for developers transitioning to Arduino IDE 2. It provides clear instructions and insights into the new features, enabling users to take full advantage of the updated development environment. This guide is essential for staying up-to-date with the latest Arduino tools[Ard24].

Literature Review Article: 21

Topic : Getting Started with Arduino mbed Core for Nano Boards I

Description: This guide provides an introduction to using the Arduino mbed core for Nano boards. It covers installation, configuration, and basic usage, enabling developers to leverage the mbed OS for advanced IoT and embedded systems projects.

Keywords: Arduino mbed Core, Nano Boards, IoT, Embedded Systems.

Topic : Getting Started with Arduino mbed Core for Nano Boards II

Conclusion: The guide is a valuable resource for developers looking to use the Arduino mbed core for advanced projects. It provides clear instructions and examples, enabling users to quickly get started with mbed OS. This guide is essential for projects requiring real-time operating systems and advanced IoT capabilities[Doc24b].

Literature Review Article: 22

Topic : Arduino Nano 33 BLE Sense Rev2 - IMU Accelerometer Tutorial I

Description: This tutorial provides a step-by-step guide to using the IMU accelerometer on the Arduino Nano 33 BLE Sense Rev2. It covers sensor initialization, data acquisition, and basic signal processing techniques for motion detection.

Keywords: IMU, Accelerometer, Arduino Nano 33 BLE Sense, Motion Detection.

Topic : Arduino Nano 33 BLE Sense Rev2 - IMU Accelerometer Tutorial II

Conclusion: The tutorial is an excellent resource for developers looking to integrate motion sensing into their projects. It provides clear explanations and practical examples, enabling users to quickly implement motion-based applications. This guide lays the foundation for more advanced projects involving IMU and accelerometer data[Doc24a].

Literature Review Article: 23

Topic : Arduino Nano 33 BLE Sense Rev2 - Humidity and Temperature Sensor Tutorial I

Description: This tutorial provides a step-by-step guide to using the humidity and temperature sensor on the Arduino Nano 33 BLE Sense Rev2. It covers sensor initialization, data acquisition, and basic signal processing techniques for environmental monitoring.

Keywords: Humidity Sensor, Temperature Sensor, Arduino Nano 33 BLE Sense, Environmental Monitoring.

Topic : Arduino Nano 33 BLE Sense Rev2 - Humidity and Temperature Sensor Tutorial II

Conclusion: The tutorial is a valuable resource for developers looking to integrate environmental sensing into their projects. It provides clear explanations and practical examples, enabling users to quickly implement humidity and temperature monitoring applications. This guide is essential for projects involving environmental data acquisition[Ard24d].

Literature Review Article: 24

Topic : Arduino Nano 33 BLE Sense Rev2 - Full Pinout Diagram I

Description: This document provides the full pinout diagram for the Arduino Nano 33 BLE Sense Rev2, detailing the functions of each pin and its connectivity options. It serves as a quick reference for hardware designers and developers.

Keywords: Arduino Nano 33 BLE Sense, Pinout, Hardware Design, Connectivity.

Topic : Arduino Nano 33 BLE Sense Rev2 - Full Pinout Diagram II

Conclusion: The pinout diagram is an essential resource for developers working with the Arduino Nano 33 BLE Sense Rev2. It simplifies the process of designing circuits and interfacing with external components, ensuring efficient and error-free hardware development[Ard24c].

Literature Review Article: 25

Topic : APDS-9960 Digital Proximity, Ambient Light, RGB and Gesture Sensor I

Description: This datasheet provides detailed information on the APDS-9960 sensor, including its features, specifications, and applications. It covers proximity detection, ambient light sensing, RGB color sensing, and gesture recognition capabilities.

Keywords: APDS-9960, Proximity Sensor, Ambient Light Sensor, RGB Sensor, Gesture Recognition.

Topic : APDS-9960 Digital Proximity, Ambient Light, RGB and Gesture Sensor II

Conclusion: The datasheet is a comprehensive resource for understanding and using the APDS-9960 sensor. It provides detailed technical information and application examples, enabling developers to integrate advanced sensing capabilities into their projects. This sensor is ideal for applications requiring proximity, light, color, and gesture detection[Tec15].

Literature Review Article: 26

Topic : Connecting Nano 33 BLE Devices over Bluetooth I

Description: This tutorial provides a step-by-step guide to connecting Arduino Nano 33 BLE devices over Bluetooth. It covers Bluetooth Low Energy (BLE) communication, device pairing, and data exchange between devices.

Keywords: Bluetooth, BLE, Arduino Nano 33 BLE, Device Communication.

Topic : Connecting Nano 33 BLE Devices over Bluetooth II

Conclusion: The tutorial is a valuable resource for developers looking to implement Bluetooth communication in their projects. It provides clear explanations and practical examples, enabling users to quickly set up and use BLE for device-to-device communication. This guide is essential for IoT and wearable technology projects[Bag22].

Literature Review Article: 27

Topic : Gait Analysis Using a Shoe-Integrated Wireless Sensor System I

Description: This study presents a wireless sensor system integrated into shoes for gait analysis. It uses inertial sensors to measure gait parameters and provides insights into normal and pathological gait patterns.

Keywords: Gait Analysis, Wireless Sensors, Inertial Sensors, Pathological Gait.

Topic : Gait Analysis Using a Shoe-Integrated Wireless Sensor System II

Conclusion: The study demonstrates the effectiveness of a shoe-integrated wireless sensor system for gait analysis. It provides valuable data for understanding gait patterns and diagnosing gait disorders. This system is a promising tool for healthcare applications, particularly in monitoring and rehabilitation[Bam+08].

Literature Review Article: 28

Topic : Inertial Measurement Unit (IMU) Sensors I

Description: This article provides an overview of IMU sensors, including their components (accelerometer, gyroscope, and magnetometer) and applications in motion sensing and navigation systems.

Keywords: IMU, Accelerometer, Gyroscope, Magnetometer, Motion Sensing.

Topic : Inertial Measurement Unit (IMU) Sensors II

Conclusion: The article is a valuable resource for understanding IMU sensors and their applications. It highlights the importance of IMUs in modern navigation and motion sensing systems, providing insights into their design and functionality. This knowledge is essential for developing advanced motion-based applications[Sol24].

Literature Review Article: 29

Topic : BMI270 Data Sheet I

Description: This datasheet provides detailed technical specifications for the BMI270 IMU sensor, including its features, performance characteristics, and application examples.

Keywords: BMI270, IMU, Accelerometer, Gyroscope, Motion Sensing.

Topic : BMI270 Data Sheet II

Conclusion: The datasheet is a comprehensive resource for developers working with the BMI270 sensor. It provides detailed information on the sensor's capabilities and performance, enabling efficient integration into motion sensing and navigation systems. This sensor is ideal for applications requiring high-precision motion detection[Sen24a].

Literature Review Article: 30

Topic : BMM150 Data Sheet I

Description: This datasheet provides detailed technical specifications for the BMM150 magnetometer, including its features, performance characteristics, and application examples.

Keywords: BMM150, Magnetometer, Magnetic Field Sensing, Navigation.

Topic : BMM150 Data Sheet II

Conclusion: The datasheet is a valuable resource for developers working with the BMM150 magnetometer. It provides detailed information on the sensor's capabilities and performance, enabling efficient integration into navigation and orientation systems. This sensor is ideal for applications requiring precise magnetic field detection[Sen24c].

Literature Review Article: 31

Topic : BMI270 Data Sheet I

Description: This datasheet provides detailed technical specifications for the BMI270 IMU sensor, including its features, performance characteristics, and application examples.

Keywords: BMI270, IMU, Accelerometer, Gyroscope, Motion Sensing.

Topic : BMI270 Data Sheet II

Conclusion: The datasheet is a comprehensive resource for developers working with the BMI270 sensor. It provides detailed information on the sensor's capabilities and performance, enabling efficient integration into motion sensing and navigation systems. This sensor is ideal for applications requiring high-precision motion detection[Sen24b].

Literature Review Article: 32

Topic : Electronic Signals and Systems: Analysis, Design and Applications I

Description: This book provides a comprehensive introduction to electronic signals and systems, covering analysis, design, and practical applications. It is a valuable resource for students and professionals in electronics and signal processing.

Keywords: Electronic Signals, Signal Processing, System Design, Applications.

Topic : Electronic Signals and Systems: Analysis, Design and Applications II

Conclusion: The book is an essential resource for understanding electronic signals and systems. It provides clear explanations and practical examples, making it accessible for beginners and experienced professionals alike. This book is a must-read for anyone working in electronics and signal processing[KO20].

Literature Review Article: 33

Topic : What are the Key Pros and Cons of the Arduino Programming Language? I

Description: This article discusses the advantages and disadvantages of the Arduino programming language, highlighting its simplicity, versatility, and limitations for advanced applications.

Keywords: Arduino, Programming Language, Pros and Cons, Embedded Systems.

Topic : What are the Key Pros and Cons of the Arduino Programming Language? II

Conclusion: The article provides a balanced overview of the Arduino programming language, making it a valuable resource for developers considering its use in their projects. It highlights the language's strengths and weaknesses, enabling informed decision-making for project development[Cha23].

Literature Review Article: 34

Topic : TensorFlow Lite Micro: Embedded Machine Learning on TinyML Systems I

Description: This paper introduces TensorFlow Lite Micro (TFLM), a framework for running machine learning models on resource-constrained embedded systems. It discusses the framework's design, implementation, and performance evaluation.

Keywords: TensorFlow Lite Micro, TinyML, Embedded Machine Learning, Inference.

Topic : TensorFlow Lite Micro: Embedded Machine Learning on TinyML Systems II

Conclusion: The paper demonstrates the effectiveness of TFLM for deploying machine learning models on embedded systems. It provides valuable insights into the framework's design and performance, making it a key resource for developers working on TinyML applications. This framework is essential for enabling AI on low-power devices[Dav+21].

Literature Review Article: 35

Topic : Factors Influencing Gait Variability and Its Implications for Mobility Disorders I

Description: This study explores the factors influencing gait variability and their implications for mobility disorders. It provides insights into the relationship between gait patterns and neurological conditions.

Keywords: Gait Variability, Mobility Disorders, Neurological Conditions, Gait Analysis.

Topic : Factors Influencing Gait Variability and Its Implications for Mobility Disorders II

Conclusion: The study highlights the importance of gait variability as a diagnostic tool for mobility disorders. It provides valuable data for understanding the relationship between gait patterns and neurological conditions, enabling better diagnosis and treatment. This research contributes to the field of gait analysis and rehabilitation[Dav20].

Literature Review Article: 36

Topic : Handbook of Modern Sensors: Physics, Designs, and Applications I

Description: This book provides a comprehensive overview of modern sensor technologies, covering their physics, design principles, and applications in various fields.

Keywords: Sensors, Physics, Design, Applications.

Topic : Handbook of Modern Sensors: Physics, Designs, and Applications II

Conclusion: The book is an essential resource for understanding modern sensor technologies. It provides detailed explanations and practical examples, making it accessible for students and professionals alike. This book is a must-read for anyone working in sensor design and applications[Fra16].

Literature Review Article: 37

Topic : GitHub Desktop: A Seamless Way to Contribute to Repositories I

Description: GitHub Desktop is a user-friendly application that simplifies the process of contributing to repositories. It provides a graphical interface for version control, making it easier for developers to manage their code and collaborate on projects.

Keywords: GitHub Desktop, Version Control, Collaboration, Repository Management.

Topic : GitHub Desktop: A Seamless Way to Contribute to Repositories II

Conclusion: GitHub Desktop is a valuable tool for developers looking to streamline their workflow and contribute to repositories more efficiently. Its intuitive interface and powerful features make it an essential tool for both beginners and experienced developers. This application enhances collaboration and productivity in software development projects[Git24].

Literature Review Article: 38

Topic : Gait Dynamics in Parkinson's Disease: Common and Distinct Behavior Among Stride Length, Gait Variability, and Fractal-like Scaling I

Description: This study investigates the gait dynamics of Parkinson's disease patients, focusing on stride length, gait variability, and fractal-like scaling. It provides insights into the unique gait patterns associated with Parkinson's disease and their implications for diagnosis and treatment.

Keywords: Gait Dynamics, Parkinson's Disease, Stride Length, Gait Variability, Fractal-like Scaling.

Topic : Gait Dynamics in Parkinson's Disease: Common and Distinct Behavior Among Stride Length, Gait Variability, and Fractal-like Scaling II

Conclusion: The study highlights the distinct gait patterns in Parkinson's disease patients, providing valuable data for understanding the disease's progression and developing targeted interventions. This research contributes to the field of gait analysis and offers new avenues for improving patient care[Hau09].

Literature Review Article: 39

Topic : Inertial Navigation System: A Survey I

Description: This paper provides a comprehensive survey of inertial navigation systems (INS), covering their components, algorithms, and applications. It discusses the challenges and advancements in INS technology, particularly in the context of autonomous navigation and robotics.

Keywords: Inertial Navigation System, INS, Autonomous Navigation, Robotics, Survey.

Topic : Inertial Navigation System: A Survey II

Conclusion: The survey highlights the importance of INS in modern navigation systems and robotics. It provides a detailed overview of the technology's evolution, challenges, and future directions, making it a valuable resource for researchers and engineers working in the field of autonomous systems[Ja18].

Literature Review Article: 40

Topic : Human Motion Activity Recognition and Pattern Analysis Using Compressed Deep Neural Networks I

Description: This study explores the use of compressed deep neural networks for human motion activity recognition and pattern analysis. It focuses on optimizing the performance of deep learning models for resource-constrained devices, enabling real-time motion analysis.

Keywords: Human Motion, Activity Recognition, Deep Neural Networks, Pattern Analysis, Compression.

Topic : Human Motion Activity Recognition and Pattern Analysis Using Compressed Deep Neural Networks II

Conclusion: The study demonstrates the effectiveness of compressed deep neural networks for human motion activity recognition. It provides valuable insights into optimizing deep learning models for low-power devices, making it a key resource for developing real-time motion analysis systems. This research contributes to the field of wearable technology and healthcare applications[Kum+24].

Literature Review Article: 41

Topic : A New Approach to Linear Filtering and Prediction Problems I

Description: This paper introduces the Kalman filter, a mathematical algorithm for linear filtering and prediction. It provides a detailed explanation of the filter's principles, applications, and advantages in various fields, including navigation and control systems.

Keywords: Kalman Filter, Linear Filtering, Prediction, Navigation, Control Systems.

Topic : A New Approach to Linear Filtering and Prediction Problems II

Conclusion: The Kalman filter is a powerful tool for solving linear filtering and prediction problems. Its applications in navigation, control systems, and signal processing have revolutionized these fields. This paper lays the foundation for further research and development of advanced filtering techniques[Kal12].

Literature Review Article: 42

Topic : Recent Advances in MEMS Sensor Technology-Mechanical Applications I

Description: This article reviews recent advancements in Micro-Electro-Mechanical Systems (MEMS) sensor technology, focusing on their mechanical applications. It discusses the design, fabrication, and performance of MEMS sensors in various industries.

Keywords: MEMS, Sensor Technology, Mechanical Applications, Design, Fabrication.

Topic : Recent Advances in MEMS Sensor Technology-Mechanical Applications II

Conclusion: The article highlights the significant progress in MEMS sensor technology and its impact on mechanical applications. It provides valuable insights into the design and performance of MEMS sensors, making it a key resource for researchers and engineers working in this field. This technology has the potential to revolutionize various industries, including automotive, aerospace, and healthcare[KS12].

Literature Review Article: 43

Topic : An Efficient Orientation Filter for Inertial and MARG Sensors I

Description: This paper presents the Madgwick filter, an efficient orientation filter for inertial and Magnetic, Angular Rate, and Gravity (MARG) sensors. It provides a detailed explanation of the filter's algorithm, performance, and applications in motion tracking and navigation.

Keywords: Madgwick Filter, Orientation Filter, Inertial Sensors, MARG Sensors, Motion Tracking.

Topic : An Efficient Orientation Filter for Inertial and MARG Sensors II

Conclusion: The Madgwick filter is a highly efficient and accurate orientation filter for inertial and MARG sensors.

University of Applied Sciences
HOCHSCHULE EMDEN·LEER

Thanks a lot

for your attention

References

References I

[AA23] Nurul Atikah Abbas und Mohd Ridzuan Ahmad. "Keyword Spotting System with Nano 33 BLE Sense Using Embedded Machine Learning Approach". In: *Jurnal Teknologi* 85.3 (2023), S. 175–182. DOI: 10.11113/jurnalteknologi.v85.18744. URL: https://journals.utm.my/jurnalteknologi.

[Ada25] Adafruit. *Adafruit LSM9DS1 Library*. Accessed: 2025-01-26. 2025. URL: https://github.com/adafruit/Adafruit_LSM9DS1.

[Alu24] Nefeli Alushi. *Getting Started with IMU and Accelerometer on Arduino Nano 33 BLE Sense*. Last revision: 17/07/2024, Accessed: 2025-01-11. 2024. URL: https://docs.arduino.cc/tutorials/nano-33-ble-sense/imu-accelerometer/.

References II

[Ard] *The Arduino Software (IDE) - Starting Guide*. Accessed: 2024-11-27. 2024. URL: https://docs.arduino.cc/learn/starting-guide/the-arduino-software-ide/.

[Ard24] Arduino. *Getting Started with Arduino IDE 2: Downloading and Installing*. Accessed: 2024-11-27. 2024. URL: https://docs.arduino.cc/software/ide-v2/tutorials/getting-started/ide-v2-downloading-and-installing/.

[Ard24] Arduino. *The Arduino Software (IDE)*. Accessed: 2024-11-27. 2024. URL: https://docs.arduino.cc/learn/starting-guide/the-arduino-software-ide/.

References III

[Ard24a] Arduino. *Arduino Nano 33 BLE Sense Rev2 - Barometric Sensor Tutorial*. Accessed: 2024-11-16. 2024. URL: https://docs.arduino.cc/tutorials/nano-33-ble-sense-rev2/barometric-sensor/.

[Ard24a] Arduino. *Introduction to PWM on Arduino*. Accessed: 2024-11-27. 2024. URL: https://docs.arduino.cc/learn/microcontrollers/analog-output/.

[Ard24a] Arduino. *UART Communication on Arduino*. Accessed: 2024-11-27. 2024. URL: https://docs.arduino.cc/learn/communication/uart.

[Ard24b] Arduino. *Arduino Nano 33 BLE Sense Rev2 - Cheat Sheet*. Accessed: 2024-11-16. 2024. URL: https://docs.arduino.cc/tutorials/nano-33-ble-sense-rev2/cheat-sheet/.

References IV

[Ard24b] Arduino. *SPI Communication with Arduino*. Accessed: 2024-11-27. 2024. URL: https://docs.arduino.cc/learn/communication/spi/.

[Ard24b] Arduino. *Understanding I2C Protocol with Arduino*. Accessed: 2024-11-27. 2024. URL: https://docs.arduino.cc/learn/communication/wire/.

[Ard24c] Arduino. *Arduino Nano 33 BLE Sense Rev2 - Full Pinout Diagram*. Accessed: 2024-11-27. 2024. URL: https://docs.arduino.cc/resources/pinouts/ABX00069-full-pinout.pdf.

[Ard24d] Arduino. *Arduino Nano 33 BLE Sense Rev2 - Humidity and Temperature Sensor Tutorial*. Accessed: 2024-11-27. 2024. URL: https://docs.arduino.cc/tutorials/nano-33-ble-sense-rev2/humidity-and-temperature-sensor/.

References V

[Ard24e] Arduino. *Arduino Nano 33 BLE Sense Rev2 Pinout*. Accessed: 2024-11-27. 2024. URL: https://docs.arduino.cc/resources/pinouts/ABX00069-full-pinout.pdf.

[Ard25] Arduino. *Arduino Guide*. Accessed: 2025-01-09. 2025. URL: https://www.arduino.cc/en/Guide.

[Bag22] José Bagur. *Connecting Nano 33 BLE Devices over Bluetooth®*. Last revision: 17/03/2022, Accessed: 2025-01-11. 2022. URL: https://docs.arduino.cc/tutorials/nano-33-ble-sense/ble-device-to-device/.

[Bam+08] Stacy J. Morris Bamberg u. a. "Gait Analysis Using a Shoe-Integrated Wireless Sensor System". In: *IEEE Transactions on Information Technology in Biomedicine* 12.4 (2008), S. 413–423. DOI: 10.1109/TITB.2007.899493.

References VI

[Cha23] Sanmit Chatterjee. *What are the Key Pros and Cons of the Arduino Programming Language?* Last updated: 26 February 2024, Accessed: 2025-01-11. 2023. URL: https://emeritus.org/blog/coding-arduino-programming-language/.

[Com23] Arduino Community. *A Difference Between Arduino Nano 33 BLE Sense vs Sense Lite*. Accessed: 2025-01-11. 2023. URL: https://forum.arduino.cc/t/a-difference-between-a-n-33-ble-sense-vs-sense-lite/1030305.

[Dav20] M. David. "Factors influencing gait variability and its implications for mobility disorders.". In: *Journal of Biomechanics* 53.8 (2020), S. 187–194.

References VII

[Dav+21] Robert David u. a. "TensorFlow Lite Micro: Embedded Machine Learning on TinyML Systems". In: *Proceedings of the 4th MLSys Conference*. MLSys. San Jose, CA, USA, 2021.

[Doc24a] Arduino Documentation. *Arduino Nano 33 BLE Sense Rev2 - IMU Accelerometer Tutorial*. Accessed: 2024-11-28. 2024. URL: https://docs.arduino.cc/tutorials/nano-33-ble-sense-rev2/imu-accelerometer/#the-bmi270_bmm150-library.

[Doc24b] Arduino Documentation. *Getting Started with Arduino mbed Core for Nano Boards*. Accessed: 2024-11-27. 2024. URL: https://docs.arduino.cc/software/ide-v1/tutorials/getting-started/cores/arduino-mbed_nano/.

References VIII

[Doc25] Arduino Documentation. *Arduino Nano 33 BLE Sense - Technical Reference and Overview*. Accessed: 2025-01-11. 2025. URL: https://docs.arduino.cc/hardware/nano-33-ble-sense/.

[Fra16] Jacob Fraden. *Handbook of Modern Sensors: Physics, Designs, and Applications*. 5th. Springer, 2016. ISBN: 978-3-319-19302-1.

[Git24] GitHub, Inc. *GitHub Desktop: A Seamless Way to Contribute to Repositories*. Accessed: 29 January 2025. 2024. URL: https://desktop.github.com/.

[Hau09] Jeffrey M Hausdorff. "Gait Dynamics in Parkinson's Disease: Common and Distinct Behavior Among Stride Length, Gait Variability, and Fractal-like Scaling". In: *Chaos: An Interdisciplinary Journal of Nonlinear Science* 19.2 (2009), S. 026113. DOI: 10.1063/1.3147408.

References IX

[Ja18] W. Jung und et al. "Inertial Navigation System: A Survey". In: *Sensors* (2018). URL: https://doi.org/10.3390/s18072216.

[Kal12] R.E. Kalman. "A new approach to linear filtering and prediction problems.". In: *Journal of Basic Engineering* 82.1 (2012), S. 35–45.

[KO20] Muhammad Nasir Khan und Others. *Electronic Signals and Systems: Analysis, Design and Applications*. 1st. River Publishers Series in Signal, Image and Speech Processing. 2020.

[KS12] Farbod Khoshnoud und Clarence de Silva. "Recent Advances in MEMS Sensor Technology-Mechanical Applications". In: *IEEE Instrumentation & Measurement Magazine - IEEE INSTRUM MEAS MAG* 15 (Apr. 2012), S. 14–24. DOI: 10.1109/MIM.2012.6174574.

References X

[Kum+24] Navita Kumari u. a. "Human Motion Activity Recognition and Pattern Analysis Using Compressed Deep Neural Networks". In: *Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization* 12.1 (2024), S. 2331052. DOI: 10.1080/21681163.2024.2331052. URL: https://doi.org/10.1080/21681163.2024.2331052.

[Libnd] Arduino Libraries. *Arduino_LSM9DS1 Library Documentation*. Retrieved from Arduino Libraries GitHub repository. n.d. URL: https://github.com/arduino-libraries/Arduino_LSM9DS1.

[Sen24a] Bosch Sensortec. *BMI270 Data Sheet*. Accessed: 2024-11-28. 2024. URL: https://docs.arduino.cc/resources/datasheets/bst-bmi270-ds000.pdf.

References XI

[Sen24b] Bosch Sensortec. *BMI270 Data Sheet*. Accessed: 2024-11-28. 2024. URL: https://docs.arduino.cc/resources/datasheets/bst-bmi270-ds000.pdf.

[Sen24c] Bosch Sensortec. *BMM150 Data Sheet*. Accessed: 2024-11-28. 2024. URL: https://docs.arduino.cc/resources/datasheets/bst-bmm150-ds001.pdf.

[Sol24] Bosch Mobility Solutions. *Inertial Measurement Unit (IMU) Sensors*. Accessed: 2024-11-28. 2024. URL: https://www.bosch-mobility.com/en/solutions/sensors/inertial-measurement-unit/.

References XII

[S.r24] Arduino S.r.l. *Arduino® Nano 33 BLE Sense Rev2 Product Reference Manual*. SKU: ABX00069 and ABX00070, modified 28/10/2024. Arduino S.r.l. Via Andrea Appiani 25, 20900 Monza, Italy, 2024. URL: https://www.arduino.cc/.

[STM24] STMicroelectronics. *ADC Conversion Basics*. Accessed: 2024-11-27. 2024. URL: https://www.st.com/resource/en/application_note/an2834-basic-adc-introduction-stmicroelectronics.pdf.

[Sup24] Arduino Support. *Use the built-in sensors on Nano 33 BLE Sense*. https://support.arduino.cc/hc/en-us/articles/360014654820-Use-the-built-in-sensors-on-Nano-33-BLE-Sense. Accessed: 2024-11-21. 2024.

References XIII

[Tec15] Avago Technologies. *APDS-9960 Digital Proximity, Ambient Light, RGB and Gesture Sensor*. Accessed: 2025-01-11. 2015.